

Skills Summary

Getting Every Sign Right in a Transformed Parabola

Use this reference while completing your worksheet or when studying.

Skill 1 — Which way does the bracket move it?

Rule: in $y = a(x - h)^2 + k$ the graph moves h units sideways, and h is the value that makes the bracket zero.

$(x - 3)$ gives $h = 3$: three units *right*. $(x + 3)$ gives $h = -3$: three units *left*. The bracket always reads the opposite way round — ask “what makes it zero?” instead of reading the sign.

Example: Find the x -intercepts of $y = (x - 4)^2 - 1$.

Step 1. The squared bracket is alone once the 1 moves across, so take square roots — and undo the bracket by doing the opposite of what it says.

Step 2. $(x - 4)^2 = 1$, so $x - 4 = \pm 1$, and undoing “minus four” means *adding* four: $x = 4 \pm 1$.
 $x = 3$ or $x = 5$

Try it: Find the x -intercepts of $y = (x + 1)^2 - 4$.

$x = -1 \pm 2$ or $x = -3$ or $x = 1$

(!) Watch out: the commonest error on this whole topic is answering $x = 1$ and $x = 5$ for $(x + 3)^2 = 4$. You must *subtract* 3: the answers are -1 and -5 .

Skill 2 — The minus in front, and the number outside

Order of operations decides the signs: square the bracket *first*, then multiply by a , then add k .

A negative a flips the parabola over, so the vertex becomes the highest point. A negative k lowers the graph. Neither of them ever changes what is inside the bracket.

Example: For $y = -2(x - 1)^2 + 6$, find y when $x = 3$.

Step 1. This is a substitution, so nothing needs solving — the only decision is the order, and the minus belongs to the 2, which acts after the squaring.

Step 2. $(3 - 1)^2 = 4$; then $-2 \cdot 4 = -8$; then $-8 + 6$.
 $y = -2$

Try it: For $y = -(x + 3)^2 + 10$, find y when $x = -1$.

$y = 6$

Skill 3 — Checking a transformation with a point

The check: once you have written a transformed equation, solve $y = 0$ or substitute one convenient x . If the answers do not sit symmetrically about the vertex you claimed, a sign is wrong.

The two x -intercepts are always the same distance from the axis of symmetry — that is the fastest way to catch a reversed bracket.

Example: Find the x -intercepts of $y = -(x - 3)^2 + 4$ and check them against the vertex.

Step 1. Set $y = 0$ and take roots; the point of the exercise is the check afterwards, so keep track of where the vertex is.

Step 2. $(x - 3)^2 = 4$ gives $x - 3 = \pm 2$, so $x = 3 \pm 2$. The vertex is at $x = 3$, and both answers are two units from it.

$$x = 1 \text{ or } x = 5$$

Try it: Find the x -intercepts of $y = -2(x + 1)^2 + 8$.

Check: $x = -1 \pm 2$